Recognizing Activities and Postural Transitions

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Draft: NN results complete; four further model studies reserved.

Abstract

We study recognition of 12 human activities and postural transitions from smartphone-derived HAPT features, emphasizing generalization to unseen participants and the recognition of rare transitions. Each model is discussed in one section under a shared dataset and evaluation framework. The completed neural network study improves test accuracy from 91.59% to 95.51% and macro-F1 from 0.8333 to 0.8641, although transition recognition and participant variation remain important limitations. The remaining model studies will extend the group comparison.

1 Introduction

Human activity recognition converts motion measurements into labels such as walking, sitting, and standing. Postural transitions extend this task to brief movements between sustained activities. These movements are scarce in HAPT, so a classifier can achieve high overall accuracy while still recognizing them poorly. The central question is therefore how modeling choices affect both common activities and rare transitions for people unseen during training.

The input is a supplied feature vector for one smartphone sensor window, and the output is one of 12 activity labels. The group study compares modeling choices under this shared prediction task, with one section per model covering its approach, results, and interpretation. Recording, labeling, and construction of the supplied features are credited to the dataset authors.

1.1 Context from prior work

Reyes-Ortiz et al. combine consecutive SVM predictions with heuristic filtering to recognize transitions from inertial measurements[2]. Acampora et al. investigate neural classifiers with short-term memory of preceding activity features[3]. Both approaches motivate attention to temporal context; the present MLP predicts each window without explicit memory. Their published scores are not direct numerical baselines because compatible labels, splits, and evaluation procedures have not been established.

Focal loss was introduced to reduce the influence of well-classified examples in dense object detection[4]. Here it is tested as an activity classification objective rather than assumed to benefit rare classes. The empirical contribution is the comparison and diagnosis of modeling choices, with additional classifier families to be integrated into the shared study.

2 Data and experiment design

2.1 Dataset characteristics

HAPT contains recordings from 30 volunteers aged 19–48 using a waist-mounted smartphone with accelerometer and gyroscope sensors sampled at 50 Hz. Supplied 561-dimensional feature vectors summarize 2.56-second windows with 50% overlap, using time- and frequency-domain statistics[1]. The official local split contains 7,767 windows from 21 training participants and 3,162 windows from nine test participants; their participant sets were checked to be disjoint. The supplied feature representation is described as normalized to $[-1, 1]$. Any alternative representation is identified in

the corresponding model section.

The six transition classes account for only 352 training windows (4.53%) and 166 test windows (5.25%). Standing has 1,423 training windows, compared with 23 for sit-to-stand, a ratio of about 62:1. Figure 1 shows this imbalance. Window counts also overstate the number of independent observations because neighboring windows overlap and each person contributes many measurements. These properties motivate class-wise metrics and participant-aware validation.

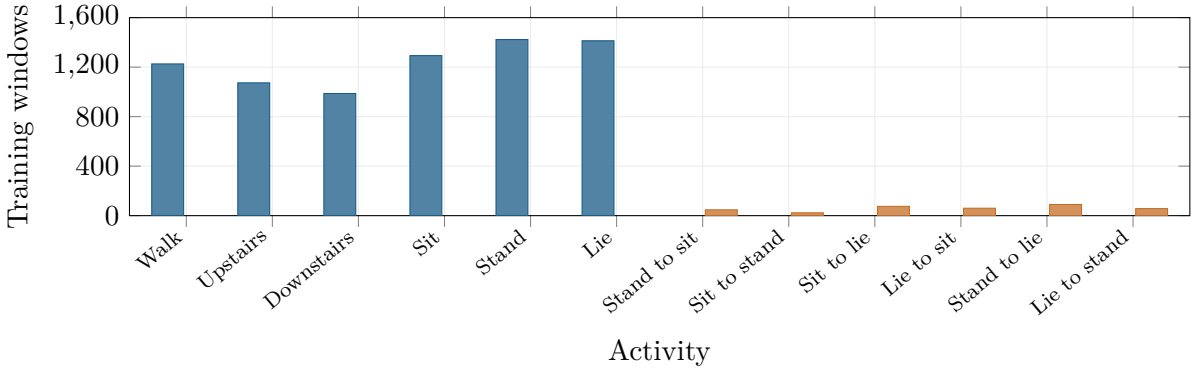

Figure 1. Training windows by class: common activities (blue) and transitions (orange). Window counts do not represent independent participants.

2.2 Selection and evaluation

The comparison is organized around the official participant-disjoint split. Tuning uses training participants only, with group-aware validation to keep correlated windows from one person within a single fold[10]. Fitted transformations such as scaling, PCA, and class weighting are learned inside the training fold. Final configurations are chosen from validation results before test evaluation. Different representations or selection procedures are identified in the relevant model section; directly comparable validation scores require matching participant folds.

Macro-F1 is the primary metric because it gives each of the 12 classes equal weight. With class counts TP_c, FP_c, and FN_c,

$$F_{1,c} = \frac{2TP_c}{2TP_c + FP_c + FN_c}, \qquad F_{1,\text{macro}} = \frac{1}{12}\sum_{c=1}^{12} F_{1,c}, \tag{1}$$

with undefined values assigned zero. Accuracy is reported alongside macro-F1, and separate means over common classes (IDs 1–6) and transitions (IDs 7–12) reveal whether an improvement reaches the minority activities. Between-fold standard deviations summarize validation variation; these tuning scores are not unbiased estimates of the full model-selection procedure.

3 Model studies

3.1 Neural network: multilayer perceptron

The NN baseline is an MLP with dimensions $561 \to 256 \to 128 \to 64 \to 12$. All hidden layers use ReLU; the first two also use batch normalization. Dropout rates are 0.3, 0.3, and 0.2[5, 6]. Hidden weights use Kaiming initialization and output weights use Xavier initialization[8, 9]. Adam uses learning rate 0.001 and batch size 128[7]. The PCA variant replaces the input with 200 training-fitted components, retaining 99.489% of training feature variance. This tests whether a smaller feature representation helps recognition without assuming that retained variance is necessarily class-relevant.

We compare cross-entropy (CE), weighted CE, focal loss, and weighted focal loss under both representations. Class weights use inverse training-class frequency; focal loss uses $\gamma = 2$ to emphasize

difficult examples[4]. Two weight decays (10^{-4}, 10^{-3}) give 16 configurations, evaluated with three participant-grouped folds and seed 42. Fold-local PCA and weights avoid validation leakage. Validation macro-F1 selects epochs and weight decay, with at most 100 epochs and early-stopping patience 12. Final refits use median selected fold epochs: 16 for the original CE baseline and 41 for the validation-selected PCA–focal variant, both with weight decay 10^{-4}.

Table 1 reports the baseline and selected test results. PCA–focal improves macro-F1 by 0.0308 and accuracy by 3.92 percentage points. However, common-activity macro-F1 rises from 0.9272 to 0.9674, while transition macro-F1 rises only from 0.7395 to 0.7608. Of the 124 fewer errors, 120 concern common activities. PCA–weighted CE has higher test macro-F1 (0.8844) and transition macro-F1 (0.8131); the validation-selected variant is retained rather than changed retrospectively using test scores.

Errors remain structured: 56 of 508 sitting windows are predicted as standing, and nine of 25 lie-to-sit windows as lie-to-stand. In test window 1151 (participant 10), sitting is predicted as standing with softmax score 0.9997. Low movement energy and a gravity-axis feature closer to the standing training mean provide descriptive evidence of weak posture discrimination. They do not establish the cause of the error, and the strong score is uncalibrated.

PCA also reduces parameters from 186,572 to 94,156, confounding representation and capacity effects. A paired bootstrap over nine test participants (5,000 draws, seed 42) gives a 95% percentile interval of $[-0.0013, 0.0670]$ for the macro-F1 improvement; it includes zero and excludes retraining variability. Parameter-matched comparisons, repeated seeds, and reliable temporal context are therefore useful next experiments. Full configurations and diagnostic outputs are available in `README_NN.md`, `results_nn_v2/`, `results_nn_pca/`, and `model_report/`; historical validation scores come from saved runs and were not independently reproduced.

3.2 Model 2: [model name]

Model study to be integrated.

3.3 Model 3: [model name]

Model study to be integrated.

3.4 Model 4: [model name]

Model study to be integrated.

3.5 Model 5: [model name]

Model study to be integrated.

4 Overall comparison

Table 1 brings the model studies together using common metrics. Only the NN rows currently contain measured results; a cross-family conclusion will follow once the remaining studies are integrated.

5 Conclusion

The completed evidence shows that an improvement in overall accuracy can be concentrated in common activities while rare transitions remain difficult. The final group comparison will consider these class-level tradeoffs alongside generalization to unseen participants, rather than ranking models by accuracy alone.

Table 1. Shared model comparison. CV entries are mean macro-F1 (sample SD); dashes indicate results not yet supplied.

Model / configuration	CV macro-F1 mean (SD)	Test accuracy (%)	Test macro-F1	Transition macro-F1
NN: original CE (baseline)	0.8503 (0.0265)	91.59	0.8333	0.7395
NN: PCA–focal (selected)	0.8735 (0.0224)	95.51	0.8641	0.7608
Model 2: [name]	—	—	—	—
Model 3: [name]	—	—	—	—
Model 4: [name]	—	—	—	—
Model 5: [name]	—	—	—	—

References

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